KELLY DIANGA

Graduate Embedded Firmware & Software Engineer | Intelligent Hardware | Robotics & Mechatronics
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PROFESSIONAL SUMMARY

MSc Software Engineering student and BEng Robotics & Mechatronics graduate with hands-on experience in embedded firmware, low-level software and hardware/software integration across STM32/ARM, Raspberry Pi/Linux, ESP32 and sensor-based projects. Practical work includes C/C++, Python, bare-metal C, GPIO/ADC/PWM/UART, FSMs, embedded Linux, edge-AI prototyping, KiCad documentation, Git, testing and debugging. Seeking an Embedded Systems / IoT role where I can contribute to firmware development, hardware bring-up, and embedded software engineering while gaining industry experience in a professional engineering environment.

TECHNICAL SKILLS

Programming: C, C++, Python, MATLAB, VHDL, Java

Embedded Firmware & Platforms: STM32 Nucleo-F446RE, ARM Cortex-M, ESP32, Arduino, Raspberry Pi, CPLD, bare-metal C, register-level programming, timers, interrupts, UART, GPIO, PWM, ADC, FSMs, WFI

Embedded Linux, RTOS & Edge AI: Linux/Raspberry Pi OS, FreeRTOS (learning), OpenCV, TensorFlow Lite, real-time camera processing, local inference

Hardware/Software Integration: Sensor/actuator interfacing, KiCad/ECAD, circuit documentation, board-level prototyping, debugging, validation, Git/GitHub

Tools & Workflows: STM32CubeIDE, Git, GitHub, GitLab, Linux, Python scripting, Docker, Jenkins, CI/CD, Xilinx ISE, Agile/Scrum

Robotics & Motion: Autonomous navigation, sensor fusion, DC/stepper/servo motors, PWM motor control

Currently Learning: STM32 bare-metal programming, FreeRTOS, advanced C++, embedded debugging/profiling, FPGA and semiconductor fundamentals

EDUCATION

MSc Software Engineering *Sep 2025 - Present*

Cardiff University | Cardiff, United Kingdom

Current taught-module average: 74.5% / Distinction level; dissertation in progress

Relevant Modules: Software Engineering, DevOps (CI/CD, Docker, Jenkins, Virtualisation, Performance Monitoring), Data Manipulation & Analysis (Pandas, NumPy, SQL)

BEng (Honours) Robotics and Mechatronics Engineering *Oct 2020 - Jan 2025*

Swinburne University of Technology | Kuching, Malaysia

Upper Second Class (2:1) | GPA: 3.29/4.00 | Dean's List of Top Achievers (2021)

Relevant Modules: Embedded Microcontrollers, Digital Electronics Design, Circuits & Electronics, Robot System Design, Electrical Actuators & Sensors, Mechatronics Systems Design, Digital Signal Processing

SELECTED TECHNICAL PROJECTS

Smart Depot Safety & Access Monitoring System (STM32) ***Jun 2026 - Present***

Independent Embedded Firmware & Intelligent Hardware Project

Stack: STM32 Nucleo-F446RE, STM32CubeIDE, C, GPIO, ADC, Timers, Interrupts, PIR Sensor, LDR, Push Button, Buzzer, LEDs, KiCad/ECAD, Git/GitHub

- Developing STM32-based embedded firmware for a safety and access monitoring prototype, integrating motion sensing, ambient light detection, push-button input, LED/buzzer alerts and state-based system behaviour.
- Implementing C firmware for ADC/GPIO handling, timer/interrupt-ready control flow, and finite-state logic for day/night operation, motion-triggered alerts, manual override and test modes.
- Creating KiCad schematics and hardware notes to support board-level integration, output-driver decisions, debugging, validation and future low-power or RTOS-based iteration.
- Managing development through Git/GitHub with structured commits, peripheral test plans, debugging notes and traceable documentation for future system improvement.

Automatic Water Tank Level Control System *May 2022 - Jun 2022*

Embedded Systems Module

Stack: ARM Cortex-M (MK20D), Bare-Metal C, ADC, GPIO, FTM Timer, ISR

- Developed bare-metal C firmware for an automated water level control prototype, integrating ADC/GPIO input handling, pump actuation and interrupt-driven timing on ARM Cortex-M hardware.
- Configured the FlexTimer Module (FTM) in Input Capture mode and directly manipulated peripheral registers (SIM, PORT, GPIO), supporting deterministic low-level hardware control.
- Used the Wait-For-Interrupt (WFI) instruction during idle monitoring cycles to reduce CPU activity, demonstrating awareness of low-power embedded behaviour.

AI-Powered Autonomous Human-Following Robot *Sep 2023 - Apr 2024*

BEng Final Year Project

Stack: Raspberry Pi 4, ESP32, Logitech Webcam, C++ (Arduino), Python, OpenCV, TensorFlow Lite, Linux (Raspbian), UART, PWM, GPIO, SolidWorks, L298N Motor Driver

- Developed embedded Linux/Python software on Raspberry Pi 4 for real-time camera processing and local TensorFlow Lite inference, enabling edge-AI tracking without cloud dependency.
- Integrated Raspberry Pi vision processing with ESP32 motor-control firmware using UART communication, supporting hardware/software co-design across Linux and microcontroller systems.
- Programmed ESP32 motor-control logic in C++ using PWM speed regulation and proportional position mapping while debugging sensor placement, actuator control and chassis integration.

VHDL Traffic Light Control System *Dec 2021*

Digital Electronics Design Module

Stack: VHDL, Xilinx ISE, CoolRunner2 CPLD, FSM, Synchronous Logic

- Designed and synthesised a traffic light control system in VHDL targeting CoolRunner2 CPLDs using Xilinx ISE, achieving a clean synthesis with 0 errors and 0 warnings across three interconnected modules.
- Implemented a 10-state, 26-transition Finite State Machine with one-hot encoding, producing Moore-style traffic outputs and Mealy-style timer outputs driven by a 4-bit synchronous counter.
- Developed VHDL testbenches covering 6+ scenarios, including simultaneous vehicle and pedestrian detection, priority ordering, and input memory retention, verifying behaviour at module and top-level integration.

WORK EXPERIENCE

STEM Ambassador (Robotics & Embedded Systems) *Oct 2025 - Present*

Cardiff University | Cardiff, United Kingdom

- Facilitate hands-on robotics and coding workshops for over 200 students, helping learners engage with LEGO SPIKE, Scratch, and micro:bit activities.
- Explain technical concepts clearly, support learners during practical tasks, and strengthen communication of engineering ideas to non-specialist audiences.

Process Engineer (Industrial Placement) *Jul 2023 - Aug 2023 | Feb 2025*

Isuzu East Africa | Nairobi, Kenya

- Conducted time-motion studies across assembly lines, identifying process bottlenecks and implementing improvements that reduced average operation time by 15%.
- Collaborated with cross-functional engineering teams to improve manufacturing data management, production traceability, and quality control.
- Gained practical exposure to vehicle assembly, end-of-line quality checks, process documentation, and manufacturing support, strengthening understanding of product lifecycle and high-volume production environments.

CERTIFICATIONS & ACHIEVEMENTS

- Dean's List of Top Achievers - Swinburne University of Technology (2021)
- Dassault Systèmes Additive Manufacturing Certification (2022)